



Family Health Portal

Project Documentation – Version 1.0 | June 2025



1. Project Overview

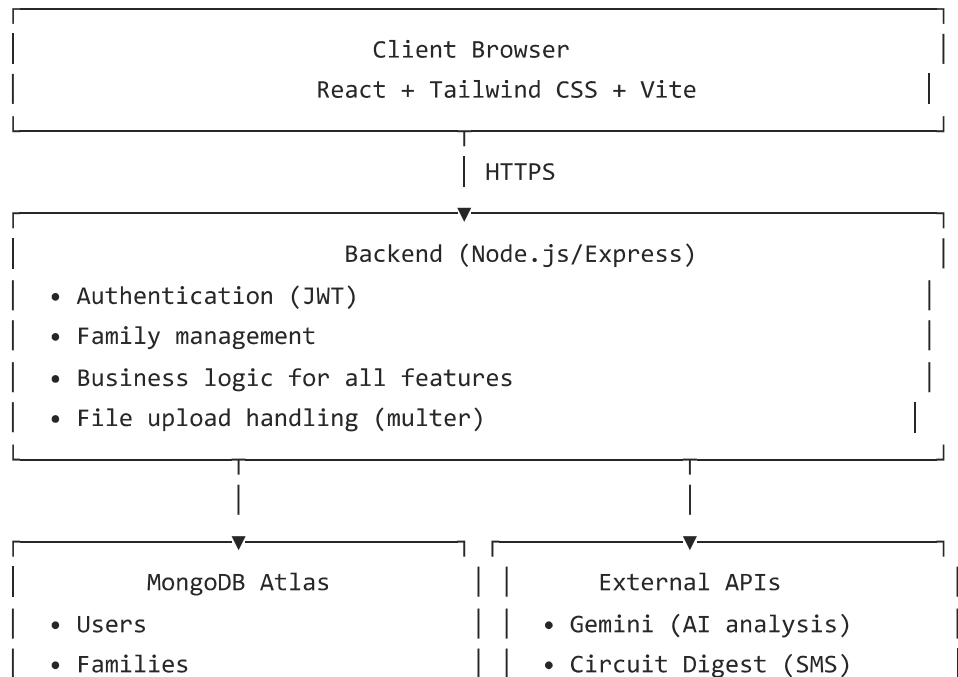
Family Health Portal is a comprehensive web application designed to help families manage their collective health data in one secure, accessible place. It allows family members to:

- Store and share health records, medications, appointments, and vital signs.
- Receive proactive reminders for medications and appointments.
- Upload medical documents, prescriptions, and reports, with automatic text extraction and AI-powered simplification.
- Trigger emergency alerts that notify all family members instantly with location data.

The system is built with a **React** frontend and **Node.js + Express + MongoDB** backend, using **JWT** authentication, role-based access (admin/member), and external APIs for notifications and AI analysis.



2. System Architecture



• HealthProfiles	• Twilio (WhatsApp/SMS)
• Medications	• Cloudinary (image storage)
• MedicationLogs	• Tesseract (OCR)
• Vitals	• pdf-parse
• Appointments	
• Prescriptions	
• HealthReports	
• MedicalDocuments	



3. Implemented Features (Detailed)

3.1 Authentication & Onboarding

- **Signup / Login:** Users register with email, password, name, and phone number (optional). Passwords hashed with bcrypt. JWT tokens manage sessions.
- **Family Creation:** The first user creates a family, generating a unique 8-character code.
- **Join Family:** Other members can join using the family code.
- **Family Roles:** Creator becomes **admin**; others are **members**. Admins can invite more members.
- **Onboarding Wizard:** Multi-step health profile (basic info, allergies, medical history, emergency contacts).

3.2 Dashboard & Navigation

- **Sidebar Navigation:** Collapsible sidebar with tabs: Dashboard, Family Members, Medications, Vitals Tracker, Appointments, Health Reports, Prescriptions, Medical Documents.
- **Family Members View:** Grid of all family members with avatars, emails, and role badges. Admin can invite new members.
- **Emergency Button:** Floating red button visible on all pages. Opens a modal that captures emergency type, custom message, and user location (Google Maps link). Sends SMS alerts to all other family members via Circuit Digest API.

3.3 Medication Manager

- **Add Medication:** Input medicine name, category, dosage, frequency (daily/custom), start/end date, time of day (morning/afternoon/evening/night).
- **Schedule Generation:** Backend automatically generates a daily schedule with exact times (10:00, 14:00, 18:00, 22:00) for the entire duration.

- **Mark as Taken:** Users mark today's doses as taken. A log entry is created in MedicationLogs.
- **Adherence Tracking:** Real-time adherence percentage (taken / scheduled). Colour-coded indicators (green $\geq 80\%$, yellow $\geq 50\%$, red $< 50\%$).
- **Explain Medicine:** Gemini API provides a simple 2-line explanation of the medicine.
- **Medication Logs:** View all historical logs for a medication (date, time, status).
- **Medication Reminders:** Cron jobs daily at 9:30, 13:30, 17:30, 21:30 (30 min before each fixed time). Sends SMS reminders via Circuit Digest to users who provided a phone number.

3.4 Vitals Tracker

- **Log Vitals:** BP (systolic/diastolic), Blood Sugar, Weight, Pulse, Temperature. Each includes value, unit, timestamp, optional notes.
- **Abnormal Detection:** Rules flag abnormal values: BP $> 140/90$, Sugar > 200 , Pulse < 60 or > 100 , Temperature < 97 or > 99.5 .
- **Chart View:** Interactive line charts (Recharts) for last 7, 30, or 90 days. Reference lines show abnormal thresholds.
- **Gemini Trend Analysis:** "Analyze Trend" sends last 10 readings to Gemini, returns plain-English summary with insights and advice.
- **Summary Cards:** Total readings, latest reading, average, abnormal count.

3.5 Appointments & Reminders

- **Create Appointment:** Enter doctor name, clinic/hospital, date, time, optional notes.
- **View/Edit/Delete:** All upcoming appointments shown in a card grid. Users can edit or cancel.
- **Reminder Cron Job:** Daily at 1:30 PM – finds appointments for next day, sends WhatsApp reminders via Twilio using a predefined template (variables: date, time).
- **Past Appointments:** Automatically hidden once date/time passes.

3.6 Prescription Upload & Conversion

- **Upload Prescription Image:** Upload photo of handwritten/printed prescription.
- **OCR Extraction:** Tesseract.js extracts raw text.
- **Gemini Structuring:** Sends extracted text to Gemini to convert into structured JSON (medicine name, dosage, frequency, duration, time of day, instructions).
- **Preview & Edit:** User sees table of extracted medicines, can edit each entry, add/remove, confirm.
- **Convert to Medications:** Confirmed medicines are created as full medication entries with schedules.

3.7 Health Report Simplifier

- **Upload Report (PDF/Image):** Any medical report.
- **Text Extraction:** PDFs via pdf-parse, images via Tesseract OCR.
- **Gemini Analysis:** Extracted text sent to Gemini with prompt: provide simple explanation (≤ 200 words) and list abnormal markers (marker, value, normal range, interpretation).
- **Store & View:** Simplified explanation and abnormal markers stored in HealthReports. Users can view, archive, or delete reports.

3.8 Medical Document Repository

- **Upload Document:** PDFs or images. Images uploaded to Cloudinary; PDFs stored locally (fallback).
- **Categorisation:** Predefined categories (prescription, lab report, discharge summary, etc.) or custom category.
- **Favourites & Search:** Mark favourite, search by filename.
- **Filter by Category:** Show only documents of a specific category.
- **View & Delete:** Images open in-app; PDFs download or open in new tab.
- **Grid View:** Cards show file icon, name, size, upload date, category badge.

3.9 Emergency Mode

- **Floating Button:** Always-visible red button (optional pulsing).
- **Modal Flow (3 steps):** (1) get user location via browser geolocation, (2) select emergency type (medical, fire, accident, security, other) + optional message, (3) success confirmation.
- **Location Capture:** Generates Google Maps link (<https://maps.google.com/?q=lat,lng>).
- **SMS Alerts:** Sends message to all other family members via Circuit Digest API – includes emergency type, user name, custom message, location link, timestamp.
- **Fallback:** If location fails, proceeds without it.



4. Technical Stack

- **Frontend:** React 18, Vite, Tailwind CSS, React Hook Form, Recharts, Axios, React Dropzone, Date-fns
- **Backend:** Node.js, Express, MongoDB (Mongoose), JWT, Bcrypt, Multer

- **AI / NLP:** Google Gemini API (custom service with model fallback: gemini-2.0-flash, gemini-2.5-flash, etc.)
- **SMS / Notifications:** Circuit Digest API (medication reminders & emergency alerts), Twilio API (appointment reminders – WhatsApp/SMS)
- **OCR & Document Processing:** Tesseract.js (images), pdf-parse (PDFs)
- **Storage:** Cloudinary (images), local filesystem (PDFs & fallback)
- **Scheduling:** Node-cron (medication reminders and appointment reminders)



5. Database Schema (Key Collections)

Collection	Purpose	Key Fields
users	User accounts	name, email, password, familyId, role, onboardingCompleted, phoneNumber
families	Family groups	familyName, familyCode, createdBy, members[]
healthprofiles	Onboarding data	userId, dateOfBirth, bloodGroup, allergies[], pastDiseases[], emergencyContacts[]
medications	Medication records	userId, familyId, medicineName, dosage, frequency, schedule[] (date,time,taken), timeOfDay[], startDate, endDate, instructions
medicationlogs	Adherence logs	medicationId, userId, scheduledTime, scheduledDate, status, takenAt
vitals	Vitals readings	userId, vitalsType, value, unit, abnormalFlag, timestamp
appointments	Appointments	userId, doctorName, clinic, dateTime, notes, reminderSent
prescriptions	Uploaded prescriptions	userId, fileName, extractedText, structuredData[], status
healthreports	Processed reports	userId, fileName, simplifiedExplanation, abnormalMarkers[]
medicaldocuments	Repository documents	userId, originalName, fileType, category, storagePath, secureUrl, isFavorite



6. API Endpoints Summary

Resource	Methods	Purpose
/api/auth	POST signup, login, GET me	Authentication
/api/family	POST create, join, GET /:familyId, GET /members	Family management
/api/onboarding	POST complete, GET profile, GET status	Health profile
/api/medications	CRUD, GET analytics, POST /:id/taken	Medication manager
/api/vitals	CRUD, GET my-chart, POST analyze-my-trend	Vitals tracker
/api/appointments	CRUD, GET my-appointments	Appointments
/api/prescriptions	POST upload, GET, POST /:id/convert	Prescription upload
/api/health-reports	POST upload, GET, PUT /:id/archive, DELETE	Health reports
/api/documents	POST upload, GET, PUT /:id/favorite, PUT /:id/category, DELETE	Medical documents
/api/emergency	POST trigger, GET history	Emergency alerts



7. Scheduled Jobs (Cron)

Job	Schedule	Description
Medication Reminders	Daily at 9:30, 13:30, 17:30, 21:30	Sends SMS via Circuit Digest for doses due in 30 minutes.
Appointment Reminders	Daily at 13:30	Finds appointments for next day, sends WhatsApp/SMS via Twilio.



9. Team & Roles

Full-stack Developer – Built entire system from scratch: frontend, backend, database, API integrations, cron jobs, and deployment.



10. Conclusion

The Family Health Portal is a production-ready application solving real-world problems for families managing chronic conditions, medication adherence, and health emergencies. Its modular design, modern tech stack, and thoughtful UX make it a strong candidate for further commercialisation or open-source contribution.

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Generated from source code documentation